The Trust-Region Subproblem with an Indefinite B: the Easy Case and the Hard Case

Worked example with figures

1 Setting

At iterate x_k the trust-region model, as defined in the lessons, is

$$m_k(p) = f_k + g_k^\top p + \tfrac{1}{2}\, p^\top B_k p, \qquad f_k = f(x_k),$$

and the subproblem is $\min_{\|p\| \le \Delta_k} m_k(p)$.

Why we drop f_k. The term f_k is a constant: it does not depend on the variable p. Adding a constant shifts every value of the function by the same amount, so it does not change the minimizer:

$$\arg\min_{\|p\| \le \Delta} \big(f_k + \tilde{m}(p)\big) = \arg\min_{\|p\| \le \Delta} \tilde{m}(p), \qquad \tilde{m}(p) := g_k^\top p + \tfrac{1}{2} p^\top B_k p.$$

Also $\nabla_p f_k = 0$, so the optimality conditions (1) below involve only g_k and B_k. Finally, in the acceptance ratio

$$\rho_k = \frac{f(x_k) - f(x_k + p_k)}{m_k(0) - m_k(p_k)}, \qquad m_k(0) - m_k(p_k) = f_k - \big(f_k + \tilde{m}(p_k)\big) = -\tilde{m}(p_k),$$

f_k cancels as well. Therefore, from now on we write

$$\min_{p \in \mathbb{R}^n} \; m(p) = g^\top p + \tfrac{1}{2}\, p^\top B p \qquad \text{s.t. } \|p\| \le \Delta,$$

with $g = g_k$, $B = B_k$ symmetric, $\Delta = \Delta_k$, and $m = \tilde{m}$. Important: every value $m(p)$ reported in this document is the predicted change of f from x_k; the model value of the lessons is $m_k(p) = f_k + m(p)$. For instance, $m(p^*) = -\frac{2}{3}$ in Example 2 means $m_k(p^*) = f_k - \frac{2}{3}$, a predicted decrease of $\frac{2}{3}$.

Write the spectral decomposition $B = Q \Lambda Q^\top$ with eigenvalues $\lambda_1 \le \lambda_2 \le \cdots \le \lambda_n$ and orthonormal eigenvectors $q_1, \ldots, q_n$.

Recall (Moré–Sorensen, Theorem 4.1 in Nocedal–Wright): p^* is a global solution iff there is $\lambda \ge 0$ with

$$(B + \lambda I) p^* = -g, \qquad \lambda(\Delta - \|p^*\|) = 0, \qquad B + \lambda I \succeq 0. \tag{1}$$

2 Why the quadratic is unbounded below when B is indefinite

Proposition 1. If $\lambda_1 < 0$, then $\inf_{p \in \mathbb{R}^n} m(p) = -\infty$.

Proof. Take $p = t\, q_1$ with $t \in \mathbb{R}$. Since $B q_1 = \lambda_1 q_1$ and $\|q_1\| = 1$,

$$m(t q_1) = t\,(g^\top q_1) + \tfrac{1}{2}\, \lambda_1 t^2.$$

This is a one-dimensional parabola opening downward ($\lambda_1 < 0$), so $m(t q_1) \to -\infty$ as $|t| \to \infty$. $\square$

So there is no unconstrained minimizer of m. Two consequences:

- The Newton point $p^N = -B^{-1}g$ (when B is nonsingular) is only a stationary point of m: $\nabla m(p^N) = 0$, but $\nabla^2 m = B$ is indefinite, so p^N is a saddle (or a maximum if $B \prec 0$). Using it as a step makes no sense.
- The constraint $\|p\| \le \Delta$ is what saves us: the ball is compact and m is continuous, so a minimizer exists (Weierstrass). It must lie on the boundary $\|p\| = \Delta$: by (1), $B + \lambda I \succeq 0$ forces $\lambda \ge -\lambda_1 > 0$, hence $\|p^*\| = \Delta$.

3 The family $p(\lambda)$

For $\lambda > -\lambda_1$ the matrix $B + \lambda I$ is positive definite, and

$$p(\lambda) = -(B + \lambda I)^{-1} g = -\sum_{j=1}^{n} \frac{q_j^\top g}{\lambda_j + \lambda}\, q_j, \qquad \|p(\lambda)\|^2 = \sum_{j=1}^{n} \frac{(q_j^\top g)^2}{(\lambda_j + \lambda)^2}.$$

On $(-\lambda_1, \infty)$ every denominator $\lambda_j + \lambda$ is positive, so differentiating term by term,

$$\frac{d}{d\lambda}\|p(\lambda)\|^2 = -2\sum_{j=1}^{n} \frac{(q_j^\top g)^2}{(\lambda_j + \lambda)^3} < 0 \quad (\text{if } g \neq 0), \qquad \lim_{\lambda \to \infty} \|p(\lambda)\|^2 = 0.$$

Hence $\lambda \mapsto \|p(\lambda)\|$ is strictly decreasing and tends to 0. Everything depends on its behaviour as $\lambda \downarrow -\lambda_1$:

- Easy case ($q_1^\top g \neq 0$): the $j = 1$ term has a pole, $\|p(\lambda)\| \to \infty$. By continuity there is a unique $\lambda^* > -\lambda_1$ with $\|p(\lambda^*)\| = \Delta$, and $p^* = p(\lambda^*)$.
- Hard case ($q_1^\top g = 0$): the pole disappears, $\|p(\lambda)\|$ stays bounded as $\lambda \downarrow -\lambda_1$, and it may happen that $\|p(\lambda)\| < \Delta$ for every admissible λ. Then no $p(\lambda)$ reaches the boundary, and we must take $\lambda = -\lambda_1$ and add a component along q_1:

$$p^* = \underbrace{-\sum_{j:\lambda_j \neq \lambda_1} \frac{q_j^\top g}{\lambda_j - \lambda_1}\, q_j}_{=:\,\hat{p}} + \tau\, q_1, \qquad \tau^2 = \Delta^2 - \|\hat{p}\|^2.$$

  Adding τq_1 does not break $(B - \lambda_1 I)p^* = -g$ because $(B - \lambda_1 I)q_1 = 0$.

Graphical explanation

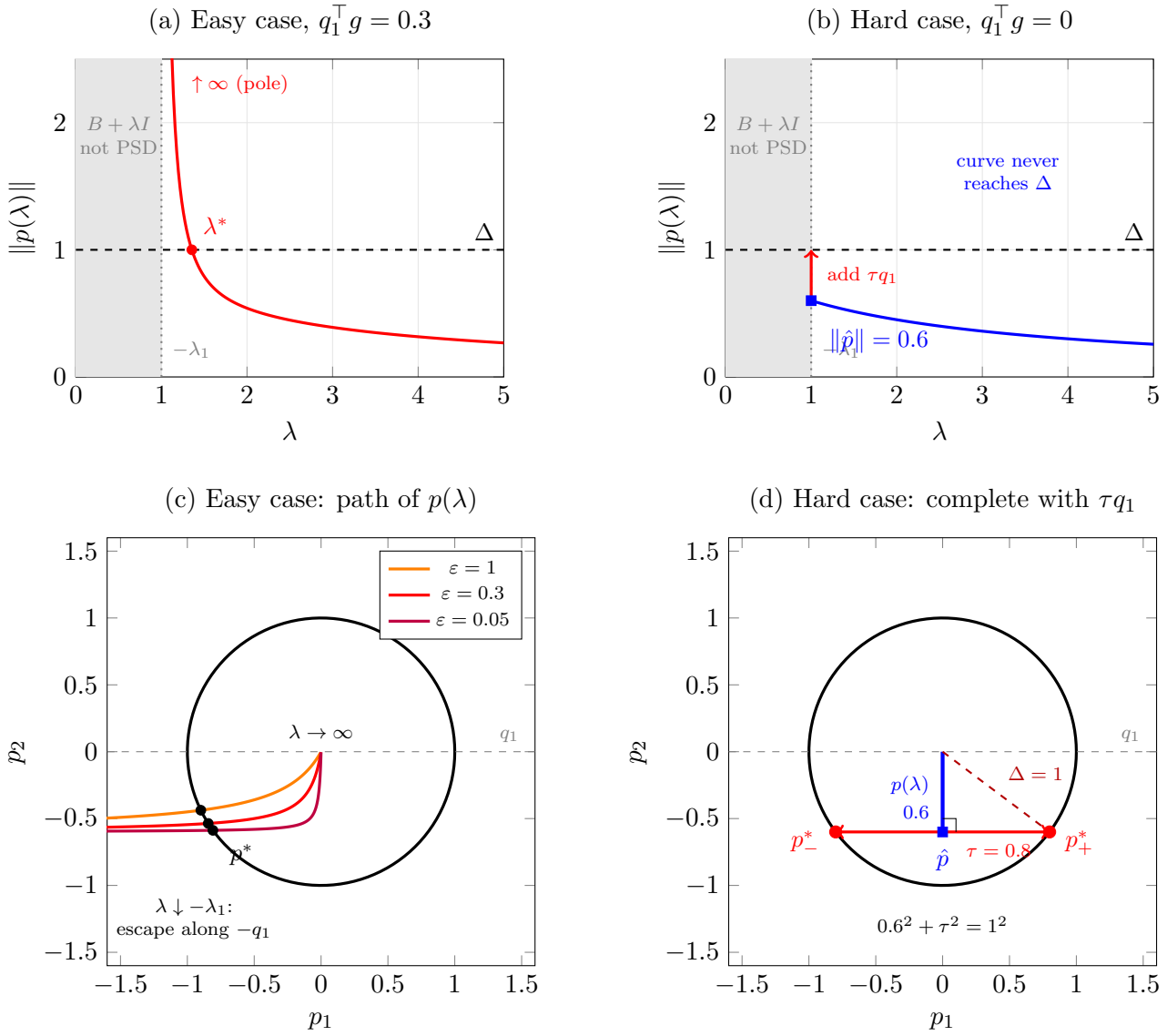


Figure 1: Easy case vs. hard case for $B = \mathrm{diag}(-1, 2)$, $\Delta = 1$, $g = (\varepsilon, 1.8)$. Top: $\|p(\lambda)\|$ as a function of λ (shaded: $\lambda < -\lambda_1$ is not admissible). Bottom: the same curves seen as paths in p-space. As $\varepsilon \to 0$, the easy-case paths (c) converge to the "L" formed by the segment $0 \to \hat{p}$ and the step $\hat{p} \to \hat{p} \pm \tau q_1$ (d).

Figure 1 illustrates both cases with $B = \mathrm{diag}(-1, 2)$, $\Delta = 1$, and $g = (\varepsilon,\, 1.8)$, so that $q_1^\top g = \varepsilon$ and

$$p(\lambda) = \left(\tfrac{-\varepsilon}{\lambda - 1},\ \tfrac{-1.8}{\lambda + 2}\right), \qquad \lambda > 1 = -\lambda_1.$$

- Top row (the scalar picture). We look for λ where the curve $\|p(\lambda)\|$ meets the line Δ. In the easy case (a) the curve starts at $+\infty$ (pole) and decreases to 0, so it must cross Δ, whatever Δ is. In the hard case (b) the curve starts at the finite value $\|\hat{p}\| = 0.6$ and decreases, so it never reaches $\Delta = 1$. The missing height $\Delta^2 - \|\hat{p}\|^2$ is supplied by the extra component τq_1 (red arrow).

- Bottom row (the picture in p-space). As λ decreases from ∞ to $-\lambda_1$, the point $p(\lambda)$ travels from the origin outward. In the easy case (c) it eventually escapes along q_1 and crosses the circle at p^*. The smaller $\varepsilon = q_1^\top g$, the longer the path hugs the p_2-axis before turning: the curves approach an "L" shape. In the limit $\varepsilon = 0$ (d) the turn never happens: the path stops at $\hat{p}$, inside the ball, and we complete the "L" by hand, moving along $\pm q_1$. Since $\hat{p} \perp q_1$, Pythagoras gives $\|\hat{p}\|^2 + \tau^2 = \Delta^2$.

So the hard case is the limit of the easy case as $q_1^\top g \to 0$: the turn of the path becomes a corner at $\hat{p}$, and the step along q_1 that the easy case produces automatically has to be added explicitly.

4 Example 1: indefinite, easy case

Example 1. Let

$$B = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \Delta = 1, \qquad m(p) = p_1 + p_2 - \tfrac{1}{2}p_1^2 + p_2^2.$$

Here $\lambda_1 = -1$, $q_1 = e_1$, $\lambda_2 = 2$, $q_2 = e_2$, and $q_1^\top g = 1 \neq 0$.

Step 1: eigen-data. B is diagonal, so its eigenvalues are its diagonal entries and its eigenvectors are the canonical vectors:

$$\lambda_1 = -1,\ q_1 = e_1 = (1, 0), \qquad \lambda_2 = 2,\ q_2 = e_2 = (0, 1), \qquad q_1^\top g = 1, \quad q_2^\top g = 1.$$

Since $\lambda_1 < 0 < \lambda_2$, B is indefinite.

Step 2: unboundedness. Along q_1: $m(t, 0) = t - \frac{1}{2}t^2 \to -\infty$; for instance $m(4, 0) = 4 - 8 = -4$ and $m(10, 0) = 10 - 50 = -40$. Along q_2: $m(0, t) = t + t^2$, a parabola opening upward. The surface is a saddle (Figure 2).

Step 3: the Newton point.

$$B^{-1} = \begin{pmatrix} -1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}, \qquad p^N = -B^{-1}g = -\begin{pmatrix} -1 \\ \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{1}{2} \end{pmatrix}, \qquad \|p^N\| = \tfrac{\sqrt{5}}{2} \approx 1.118 > \Delta.$$

Value: $m(p^N) = 1 - \frac{1}{2} - \frac{1}{2}(1)^2 + (-\frac{1}{2})^2 = \frac{1}{4}$. Since $\nabla m(p^N) = 0$, for any direction d we have exactly $m(p^N + d) = m(p^N) + \frac{1}{2}d^\top B d$. Hence

$$m(p^N + te_1) = \tfrac{1}{4} - \tfrac{1}{2}t^2 < m(p^N), \qquad m(p^N + te_2) = \tfrac{1}{4} + t^2 > m(p^N),$$

so p^N is a saddle point, not a minimizer. (It is also outside the ball.)

Step 4: admissible λ. By (1) we need $B + \lambda I \succeq 0$:

$$B + \lambda I = \begin{pmatrix} \lambda - 1 & 0 \\ 0 & \lambda + 2 \end{pmatrix} \succeq 0 \iff \lambda \geq 1 = -\lambda_1.$$

Since $\lambda \geq 1 > 0$, complementarity gives $\|p^*\| = \Delta = 1$.

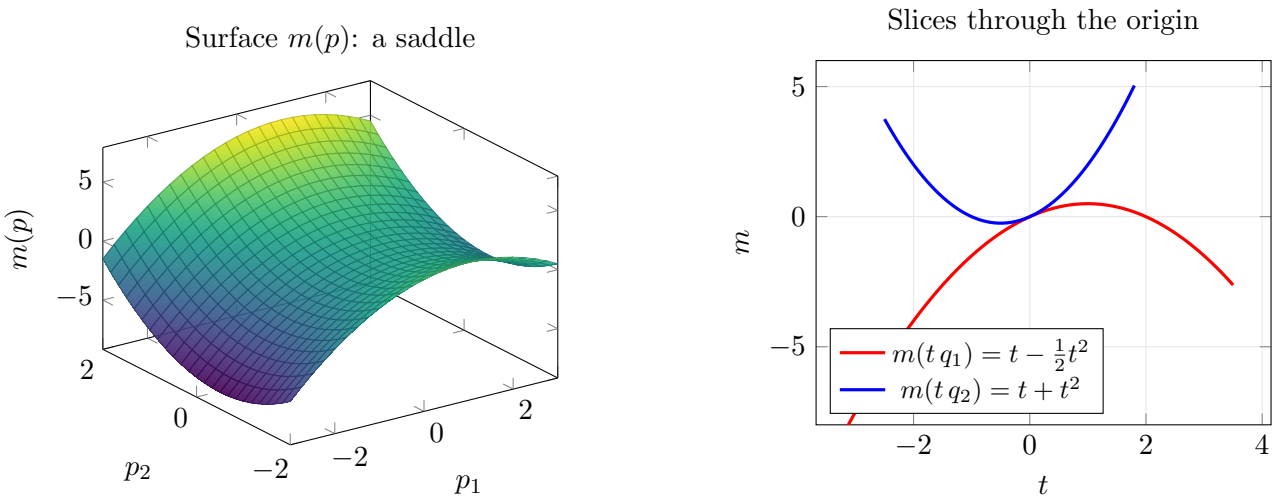


Figure 2: Left: m has negative curvature along $q_1 = e_1$ and positive curvature along $q_2 = e_2$. Right: along q_1 the model goes to $-\infty$, so it is unbounded below on $\mathbb{R}^2$.

Step 5: the curve $p(\lambda)$. For $\lambda > 1$,

$$(B + \lambda I)^{-1} = \begin{pmatrix} \frac{1}{\lambda - 1} & 0 \\ 0 & \frac{1}{\lambda + 2} \end{pmatrix}, \qquad p(\lambda) = -(B + \lambda I)^{-1} g = \left(\tfrac{-1}{\lambda - 1},\ \tfrac{-1}{\lambda + 2} \right),$$

$$\|p(\lambda)\|^2 = \frac{1}{(\lambda - 1)^2} + \frac{1}{(\lambda + 2)^2}.$$

As $\lambda \downarrow 1$ the first term blows up (pole), and as $\lambda \to \infty$ both go to 0, so the curve crosses $\Delta = 1$ exactly once (Figure 5, red).

Step 6: bracketing λ^*. A few values:

λ	1.5	2	2.5	3
$\|p(\lambda)\|$	2.020	1.031	0.703	0.539

At $\lambda = 2$: $\|p\|^2 = 1 + \frac{1}{16} = 1.0625 > 1$; at $\lambda = 3$: $\|p\|^2 = \frac{1}{4} + \frac{1}{25} = 0.29 < 1$. Hence $\lambda^* \in (2, 3)$, and close to 2.

Step 7: Newton's method for λ^*. Nocedal–Wright (Algorithm 4.3) apply Newton to the secular equation $\phi(\lambda) = \frac{1}{\Delta} - \frac{1}{\|p(\lambda)\|} = 0$, which is nearly linear in λ. With the Cholesky factor $R^\top R = B + \lambda I$ and $R^\top q = p$, the update is

$$\lambda_{\ell+1} = \lambda_\ell + \left(\frac{\|p_\ell\|}{\|q_\ell\|} \right)^2 \frac{\|p_\ell\| - \Delta}{\Delta}.$$

Here $R = \operatorname{diag}(\sqrt{\lambda - 1}, \sqrt{\lambda + 2})$, so $q = \left(p_1/\sqrt{\lambda - 1},\ p_2/\sqrt{\lambda + 2} \right)$. Starting from $\lambda_0 = 3$:

ℓ	λ_ℓ	$\|p_\ell\|$	$\|q_\ell\|$
0	3.000000	0.538516	0.364692
1	1.993758	1.036966	1.017183
2	2.032175	1.000067	0.961573
3	2.032248	1.000000	0.961473

Three iterations give $\lambda^* \approx 2.032248$.

Step 8: the solution. With $\lambda^* - 1 = 1.032248$ and $\lambda^* + 2 = 4.032248$,

$$p^* = \left(\tfrac{-1}{1.032248},\ \tfrac{-1}{4.032248}\right) \approx (-0.96876,\ -0.24800), \qquad \|p^*\|^2 \approx 0.93850 + 0.06150 = 1.$$

Value of the model, term by term:

$$g^\top p^* \approx -0.96876 - 0.24800 = -1.21676, \qquad \tfrac{1}{2} p^{*\top} B p^* \approx \tfrac{1}{2}\big(-0.93850 + 2(0.06150)\big) = -0.40774,$$
$$m(p^*) \approx -1.21676 - 0.40774 = -1.62450.$$

Step 9: checks.

- KKT (1): $(B + \lambda^* I)p^* \approx (1.032248 \cdot (-0.96876),\ 4.032248 \cdot (-0.24800)) = (-1, -1) = -g$; $\|p^*\| = \Delta$; $B + \lambda^* I = \mathrm{diag}(1.032,\ 4.032) \succ 0$.
- Comparison with other boundary points: $m(-1, 0) = -1 - \frac{1}{2} = -1.5$, $m(0, -1) = -1 + 1 = 0$, $m(1, 0) = 1 - \frac{1}{2} = 0.5$. All are larger than $m(p^*) \approx -1.6245$. A fine scan of $m(\cos\theta, \sin\theta)$ over $\theta \in [0, 2\pi)$ gives the same minimum.
- Note that p^* points mostly along $-q_1$: the solver uses the negative-curvature direction, which is exactly where m decreases fastest.

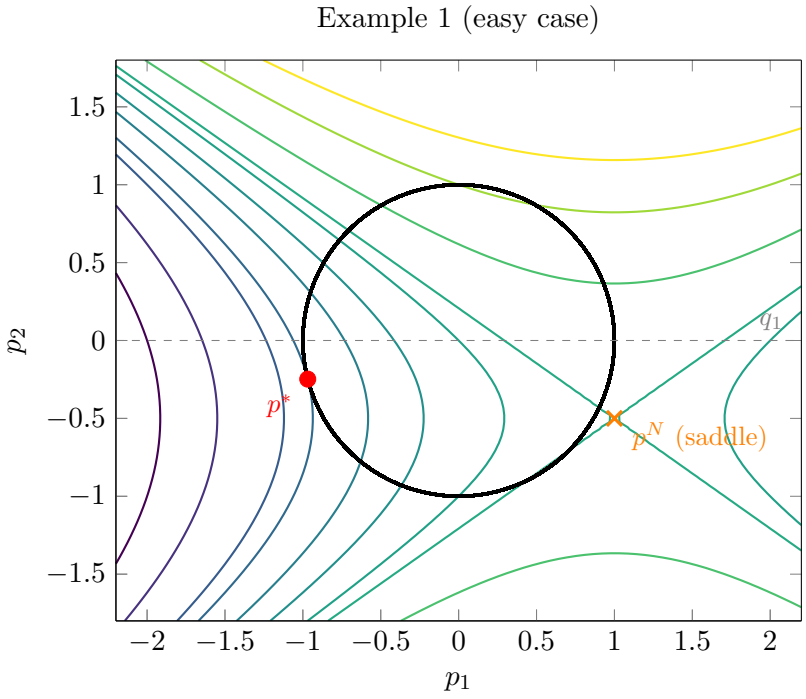


Figure 3: Level sets of m (hyperbolas, because B is indefinite), the trust region $\|p\| \le 1$ (black circle), the Newton point p^N (a saddle), and the solution p^*, where a level curve is tangent to the circle.

5 Example 2: indefinite, hard case

Example 2. Same B and $\Delta = 1$, but now

$$g = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad m(p) = p_2 - \tfrac{1}{2}p_1^2 + p_2^2.$$

Now $q_1^\top g = e_1^\top g = 0$: the gradient has no component along the direction of negative curvature.

Step 1: eigen-data. Same eigenpairs as before, but now

$$q_1^\top g = e_1^\top (0,1) = 0, \qquad q_2^\top g = 1.$$

Step 2: still unbounded. $m(t,0) = -\frac{1}{2}t^2 \to -\infty$ (e.g. $m(4,0) = -8$), so the model is unbounded below. The Newton point is

$$p^N = -B^{-1}g = -\begin{pmatrix} -1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix}, \qquad m(p^N) = -\tfrac{1}{2} + \tfrac{1}{4} = -\tfrac{1}{4},$$

and again $m(p^N + te_1) = -\frac{1}{4} - \frac{1}{2}t^2$, $m(p^N + te_2) = -\frac{1}{4} + t^2$: a saddle.

Step 3: $p(\lambda)$ cannot reach the boundary. For $\lambda > 1$,

$$p(\lambda) = -\begin{pmatrix} \frac{1}{\lambda-1} & 0 \\ 0 & \frac{1}{\lambda+2} \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \left(\tfrac{-0}{\lambda-1},\ \tfrac{-1}{\lambda+2}\right) = \left(0,\ \tfrac{-1}{\lambda+2}\right), \qquad \|p(\lambda)\| = \frac{1}{\lambda+2}.$$

The numerator $q_1^\top g = 0$ kills the term with denominator $\lambda - 1$: no pole at $\lambda = 1$. The function $\frac{1}{\lambda+2}$ is decreasing, so its supremum on $(1,\infty)$ is the limit at $\lambda = 1$:

$$\sup_{\lambda>1} \|p(\lambda)\| = \lim_{\lambda \downarrow 1} \frac{1}{\lambda+2} = \frac{1}{3} < 1 = \Delta.$$

So $\|p(\lambda)\| = \Delta$ has no solution with $\lambda > 1$ (Figure 5, blue). Note also that $p(\lambda)$ always has $p_1 = 0$; it never uses the direction q_1 along which m decreases.

Step 4: why λ^* must be exactly 1. From (1), $\lambda^* \geq 1$. If $\lambda^* > 1$, then $p^* = p(\lambda^*)$ has $\|p^*\| < \frac{1}{3} < \Delta$, and complementarity $\lambda^*(\Delta - \|p^*\|) = 0$ would force $\lambda^* = 0$, a contradiction. Hence $\lambda^* = 1$.

Step 5: solving the singular system. At $\lambda = 1$, $B + I = \mathrm{diag}(0,3)$ is singular, and

$$(B+I)p = -g \iff \begin{pmatrix} 0 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} p_1 \\ p_2 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix} \iff \begin{cases} 0 \cdot p_1 = 0 & (\text{any } p_1) \\ 3p_2 = -1 & (p_2 = -\frac{1}{3}) \end{cases}$$

The solution set is the whole line $\{(\tau, -\frac{1}{3}) : \tau \in \mathbb{R}\} = \hat{p} + \mathbb{R}\, q_1$, with $\hat{p} = (0, -\frac{1}{3})$ (the minimum-norm point, i.e. the limit of $p(\lambda)$ as $\lambda \downarrow 1$). The free parameter τ is fixed by $\|p^*\| = \Delta$:

$$\tau^2 + \tfrac{1}{9} = 1 \implies \tau^2 = \tfrac{8}{9} \implies \tau = \pm\tfrac{2\sqrt{2}}{3} \approx \pm 0.9428.$$

$$p^*_\pm = \left(\pm\tfrac{2\sqrt{2}}{3},\ -\tfrac{1}{3}\right).$$

Step 6: value of the model. Term by term:

$$g^\top p^* = -\tfrac{1}{3}, \qquad p^{*\top} B p^* = -\tau^2 + 2 \cdot \tfrac{1}{9} = -\tfrac{8}{9} + \tfrac{2}{9} = -\tfrac{6}{9} = -\tfrac{2}{3},$$

$$m(p^*) = -\tfrac{1}{3} + \tfrac{1}{2}\left(-\tfrac{2}{3}\right) = -\tfrac{1}{3} - \tfrac{1}{3} = -\tfrac{2}{3}.$$

Step 7: checks.

- KKT (1): $(B+I)p^* = \operatorname{diag}(0,3)\,(\pm\frac{2\sqrt{2}}{3}, -\frac{1}{3})^\top = (0,-1)^\top = -g$; $\|p^*\| = 1 = \Delta$; $B + I \succeq 0$.
- The best point of the family $p(\lambda)$: points $(0, s)$ with $s \in (-\frac{1}{3}, 0)$ give $m(0, s) = s + s^2$, whose infimum on that interval is at $s = -\frac{1}{3}$: $-\frac{1}{3} + \frac{1}{9} = -\frac{2}{9}$. This is worse than $-\frac{2}{3}$: moving along q_1 gains an extra $-\frac{1}{2}\tau^2 = -\frac{4}{9}$.
- Independent check on the boundary: with $p = (\cos\theta, \sin\theta)$ and $s = \sin\theta \in [-1, 1]$,

$$m = s - \tfrac{1}{2}(1 - s^2) + s^2 = \tfrac{3}{2}s^2 + s - \tfrac{1}{2}, \qquad \frac{dm}{ds} = 3s + 1 = 0 \;\Rightarrow\; s = -\tfrac{1}{3},$$

$$m\big|_{s=-1/3} = \tfrac{3}{2}\cdot\tfrac{1}{9} - \tfrac{1}{3} - \tfrac{1}{2} = \tfrac{1}{6} - \tfrac{1}{3} - \tfrac{1}{2} = -\tfrac{2}{3}, \qquad \cos\theta = \pm\sqrt{1 - \tfrac{1}{9}} = \pm\tfrac{2\sqrt{2}}{3}.$$

Same two points, same value. There are two global solutions, by symmetry (Figure 4).

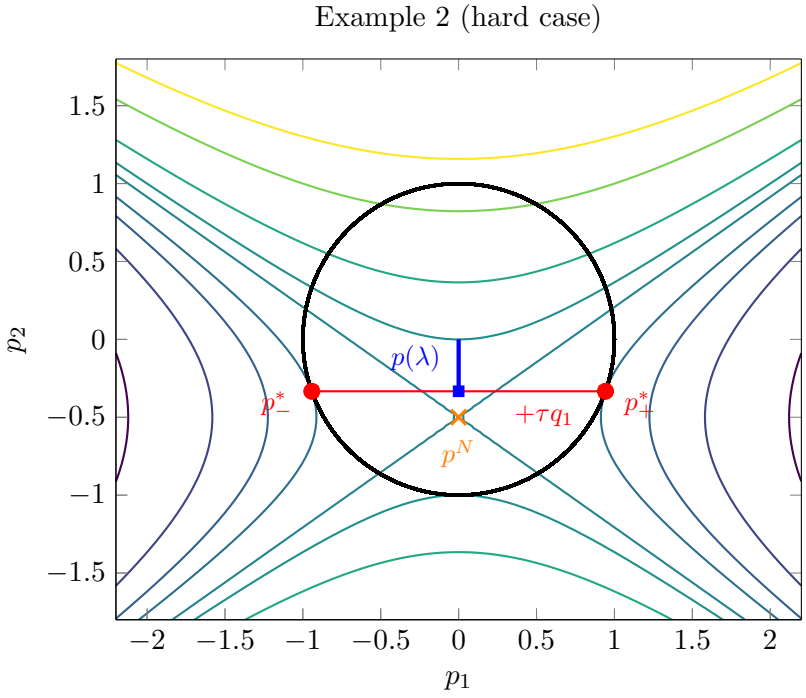


Figure 4: Hard case. The curve $p(\lambda)$, $\lambda \in (1, \infty)$, is only the short blue segment from $\hat{p} = (0, -\frac{1}{3})$ to the origin; it never touches the circle. The solutions are obtained by moving from $\hat{p}$ along $\pm q_1$ until the boundary.

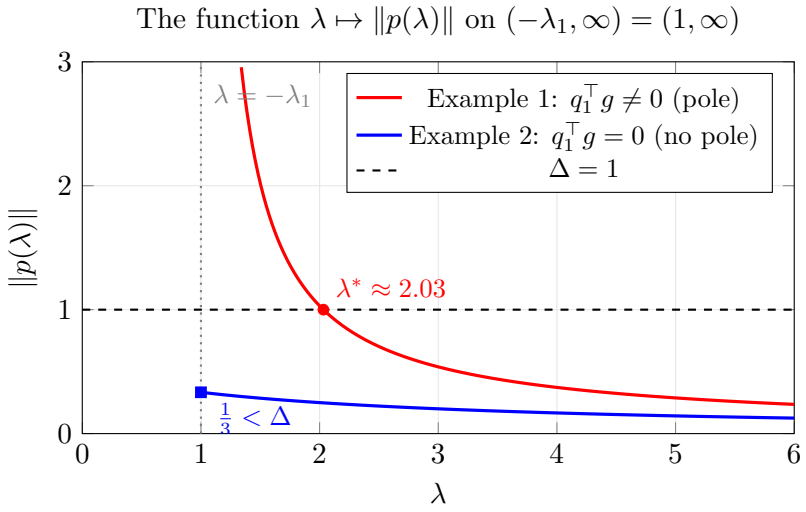


Figure 5: In the easy case the curve comes down from $+\infty$ and must cross Δ. In the hard case it is bounded by $\frac{1}{3}$, so no $\lambda > -\lambda_1$ gives $\|p(\lambda)\| = \Delta$.

6 Summary

	Example 1 (easy)	Example 2 (hard)
$q_1^\top g$	1	0
m bounded below on $\mathbb{R}^2$?	no	no
Newton point p^N	$(1, -\frac{1}{2})$, saddle	$(0, -\frac{1}{2})$, saddle
$\lim_{\lambda \downarrow 1} \|p(\lambda)\|$	$+\infty$	$\frac{1}{3} < \Delta$
λ^*	$\approx 2.032 > -\lambda_1$	$= -\lambda_1 = 1$
$B + \lambda^* I$	positive definite	PSD, singular
p^*	$p(\lambda^*)$, unique	$\hat{p} \pm \tau q_1$, two solutions
$m(p^*)$	≈ -1.6245	$-\frac{2}{3}$

Remark. The hard case requires both $q_1^\top g = 0$ and $\|\hat{p}\| < \Delta$. If $q_1^\top g = 0$ but $\Delta < \|\hat{p}\|$ (e.g. $\Delta = 0.2$ in Example 2), the blue curve does cross Δ and the problem is solved by $p(\lambda^*)$ as usual. In floating point, exact orthogonality is rare, but $q_1^\top g \approx 0$ makes the pole extremely narrow, which is why practical algorithms (Moré–Sorensen) treat this case explicitly.